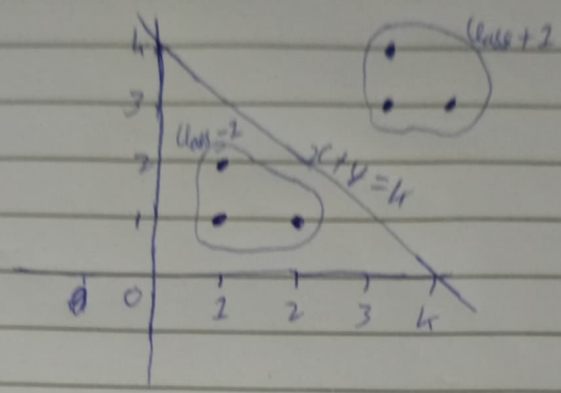
Mohd Talha Pathanwala
CMPN-B
23102B0025

ML: Assignment - 9

Q2.

(a)



(b) $x+y=4$ also gives the maximum margin.

(c) $(2,1), (1,2), (3,3)$

(d) ~~For (1,2)~~ $d_{(1,2)} = \frac{|1+2-4|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

$d_{(2,1)} = \frac{|2+1-4|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

$d_{(3,3)} = \frac{|3+3-4|}{\sqrt{2}} = \sqrt{2}$

(e) SVM chooses the maximum margin line to ensure better generalization and ~~better~~ robustness. A random separating line may lie too close to data points, making it sensitive to noise and causing ~~more~~ misclassification.

Q.2.

(a) Minimize: $\frac{1}{2} \|w\|^2$

(b) For every training sample (x_i, y_i):

$$y_i(w \cdot x_i + b) \geq 1 \quad \forall i, \quad y_i \in \{+1, -1\}$$

(c) Margin Formula:

$$\text{Margin} = \frac{2}{\|w\|}$$

-> $\text{margin} \propto \frac{1}{\|w\|}$

-> Instead of maximizing $\frac{2}{\|w\|}$, we minimize $\frac{1}{2}\|w\|^2$

(d) i) It guarantees global minimum.
ii) It is relatively simpler
iii) The solution given is equivalent to maximizing.

(e) If dataset is not linearly separable, Hard margin SVM will fail to find a solution because no perfect separating line exists.
So, Soft-margin SVM is used, allowing some errors while still maximizing the margin.

Q.3. ① For a text classification dataset with thousands of sparse features, the linear kernel is the most suitable choice because the data is high-dimensional and often already linearly separable. It is computationally efficient, scales well, and avoids unnecessary model complexity.

② For a dataset that is almost linearly separable, the linear kernel is ideal. It provides a simple, fast, and effective solution while maintaining good generalization.

③ For a dataset with highly curved and complex class boundaries, the RBF kernel is most appropriate as it can capture non-linear patterns by mapping data into higher dimensions. This allows the model to form flexible decision boundaries for complex data.